

INFO8006 Introduction to Artificial Intelligence

Exercise session 3: Quantifying uncertainty

RECAP

Kolmogorov's axioms

A *probability space* is a *sample space* Ω , whose elements ω are the *sample points*, equipped with a probability function P such that $0 \leq P(\omega) \leq 1$ and $\sum_{\omega \in \Omega} P(\omega) = 1$. An *event* is a set of sample points $A \subseteq \Omega$, of probability $P(A) = \sum_{\omega \in A} P(\omega)$. Hence $P(\Omega) = 1$, $P(A \cup B) = P(A) + P(B)$ for disjoint events, $P(\neg a) = 1 - P(a)$ and $P(a \vee b) = P(a) + P(b) - P(a \wedge b)$.

Random variables

A *random variable* is a function $X : \Omega \rightarrow D_X$, with $P(X = x) = \sum_{\{\omega : X(\omega) = x\}} P(\omega)$. A number is written $P(x)$, short for $P(X = x)$; a whole *distribution* is written $\mathbf{P}(X)$, *i.e.* the table of the values $P(X = x)$ for all $x \in D_X$.

Conditional probability, sum and product rules

For $P(b) > 0$, the *conditional probability* of a given b is $P(a | b) = P(a, b)/P(b)$, hence the *product rule* $P(a, b) = P(a | b)P(b) = P(b | a)P(a)$. Summing a joint distribution over the values of a variable marginalizes it out (*sum rule*), $P(x) = \sum_y P(x, y)$, and with the product rule it gives the *law of total probability*, $P(x) = \sum_y P(x | y)P(y)$. Selecting the entries consistent with the evidence $\mathbf{e}$, marginalizing the hidden variables out and normalizing gives the posterior over a query variable Q , $\mathbf{P}(Q | \mathbf{e}) = \frac{1}{Z} \mathbf{P}(Q, \mathbf{e})$ with $Z = \sum_q P(q, \mathbf{e})$, the constant $1/Z$ being also written α .

Bayes' rule

From the two factorizations above,

$$P(a | b) = \frac{P(b | a)P(a)}{P(b)},$$

where $P(a)$ is the prior belief on a , $P(b)$ the probability of the evidence b , $P(a | b)$ the posterior belief on a given b , and $P(b | a)$ the likelihood.

Chain rule

Any joint distribution is an incremental product of conditional distributions, $P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1, \dots, x_{i-1})$.

Independence

$A \perp B$ iff $P(a, b) = P(a)P(b)$, and $A \perp B | C$ iff $P(a, b | c) = P(a | c)P(b | c)$, for all values of the variables. Conditional independence factors a joint distribution into smaller tables, which is what makes probabilistic models scale.

In session: exercises 1, 2, 3, 5.

Exercise 1 Rational beliefs (AIMA ch. 12)

Let A and B be two *events* over the probability space Ω . An agent holds the beliefs $P(A) = 0.4$ and $P(B) = 0.3$. What ranges of probabilities would it be rational for the agent to hold for the events $A \cup B$ and $A \cap B$? What if the agent also believes that $P(A | B) = 0.5$?

Exercise 2 Full joint distribution (AIMA ch. 12)

Given the probability table below, compute the following probabilities.

	toothache		no toothache	
	catch	no catch	catch	no catch
cavity	0.108	0.012	0.072	0.008
no cavity	0.016	0.064	0.144	0.576

- (a) $P(\text{toothache})$
- (b) $P(\text{cavity})$
- (c) $P(\text{toothache} | \text{cavity})$
- (d) $P(\text{cavity} | \text{toothache} \cup \text{catch})$

Exercise 3 Rare disease (AIMA ch. 12)

After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease and that the test is 99 % accurate, *i.e.* the probability of testing positive when you do have the disease is 0.99 and the probability of testing positive when you don't is 0.01. The good news is that this is a rare disease, striking only 1 in 10 000 people of your age. Why is it good news that the disease is rare? What are the chances that you actually have the disease?

Exercise 4 Conditional independence (AIMA ch. 12)

For each of the following statements, either prove it is valid or give a counterexample.

- (a) If $P(a | b, c) = P(b | a, c)$, then $P(a | c) = P(b | c)$.
- (b) If $P(a | b, c) = P(a)$, then $P(b | c) = P(b)$.
- (c) If $P(a | b) = P(a)$, then $P(a | b, c) = P(a | c)$.

Exercise 5 Bag of coins (AIMA ch. 13)

We have a bag of three biased coins a , b , and c with probabilities of coming up heads of 20 %, 60 %, and 80 %, respectively. One coin is drawn randomly from the bag (with equal probability of drawing each of the three coins), and then the coin is flipped three times to generate the outcomes X_1 , X_2 and X_3 (heads or tail).

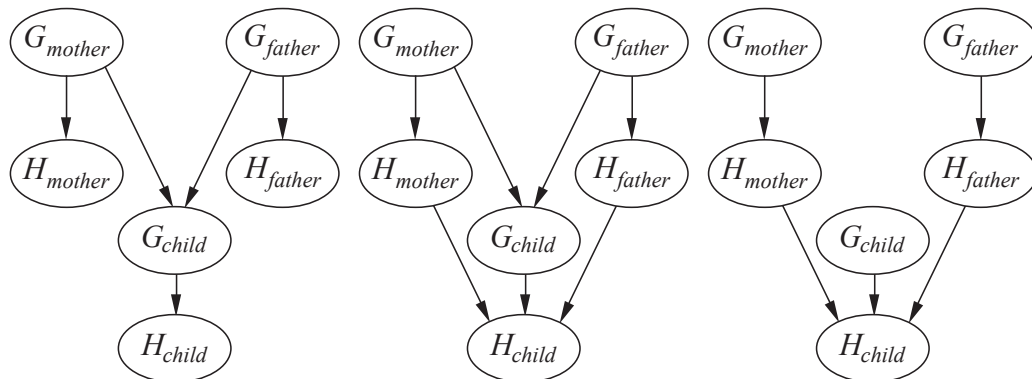
- (a) Draw the Bayesian network corresponding to this setup and define the corresponding conditional probability tables (CPTs).
- (b) Determine which coin (a , b or c) is most likely to have been drawn from the bag if the observed tosses came out heads twice and tail once.

Exercise 6 Handedness (AIMA ch. 13)

Let H_x be a random variable denoting the handedness of an individual x , with possible values l or r . A common *hypothesis* is that left or right-handedness is inherited by a simple mechanism; that is, there is a gene G_x , also with values l or r , and H_x turns out the same as G_x with some probability s .

Furthermore, the gene itself is equally likely to be inherited from either of the individual's parents, with a small nonzero probability m of a random mutation flipping the gene.

- (a) Which of the following networks claim that $P(G_{\text{child}} \mid G_{\text{mother}}, G_{\text{father}}) = P(G_{\text{child}})$?



- (b) Which of the networks make independence claims that are consistent with the hypothesis about the inheritance of handedness?
- (c) Write down the CPT for the G_{child} nodes in first network, in terms of s and m .
- (d) Suppose that $P(G_{\text{mother}} = l) = P(G_{\text{father}} = l) = q$. In the first network, derive an expression for $P(G_{\text{child}} = l)$ in terms of s , m and q .
- (e) Under conditions of genetic equilibrium, we expect the distribution of genes to be the same across generations. Use this to calculate the value of q , and, given what you know about handedness in humans, explain why the hypothesis described at the beginning of this question must be wrong.

QUIZ

1. The Bayes' rule states that ...

- ☐ $P(x \mid y) = \frac{P(y \mid x)P(y)}{P(x)}$.
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- ☐ $P(x) = \frac{P(x \mid y)P(y)}{P(x)}$.

2. Which of the following is true?

- ☐ $P(x, y \mid z) = \frac{P(x \mid y, z)}{P(y, z \mid x)}$.
- ☐ $P(x, y, z) = P(x)P(y)P(z)$.
- ☐ $P(x, y \mid z) = P(y \mid x, z)P(x \mid z)$.
- ☐ $P(x \mid y, z) = \frac{P(x, y, z)}{P(y)P(z)}$.

3. Naive Bayes model assumes ...

- ☐ conditional independence between the effects, given the cause.
- ☐ independence between the effects.
- ☐ independence between the effects and the cause.
- ☐ conditional independence between each effect and the cause, given the other effects.

4. The Bayes' rule can be read as "The posterior is equal to ...
- ☐ the product of the likelihood and the prior".
 - ☐ the ratio between the posterior distribution and the evidence, times the prior".
 - ☐ the likelihood to prior ratio multiplied by the evidence".
 - ☐ the likelihood to evidence ratio multiplied by the prior".